# RNA-Protein Interaction Classification via Sequence Embeddings

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## In a nutshell

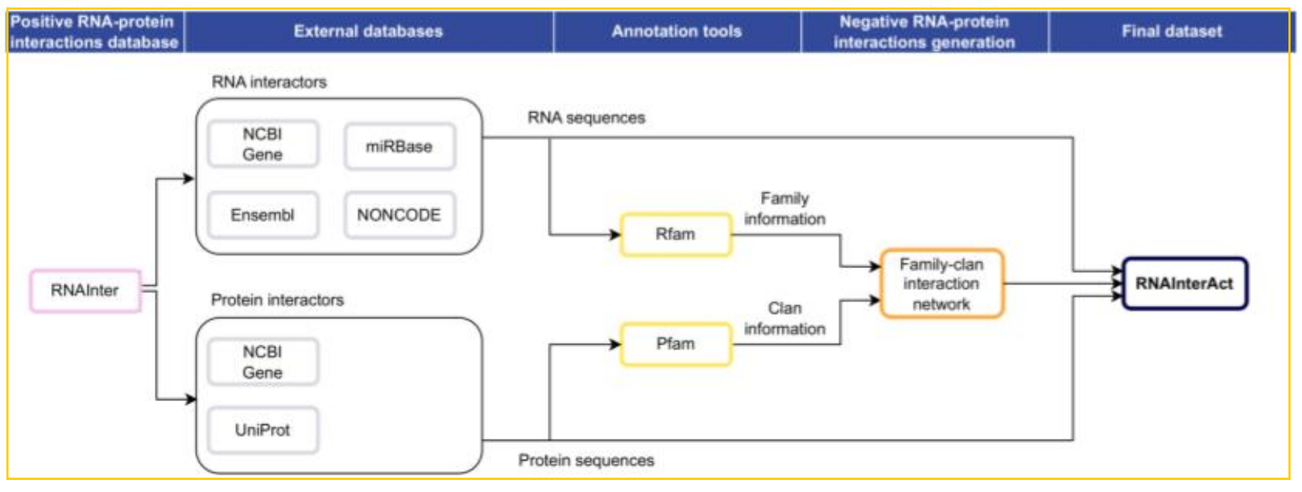
- RNA and proteins form complexes through **binding interactions**, crucial for **advances in biology or medicine**, such as drug development.
- Predicting RNA-protein interactions (RPIs) computationally is challenging due to **data scarcity** and its **bias**.
- We present **RNAInterAct,** the largest publicly available dataset with **biologically plausible negative interactions**.
- The dataset includes **sequence information** and precomputed **embeddings that capture meaningful information** about interactors' structure and function.
- A meticulous train/test split based on the **RNA family** enables evaluation on truly unseen interactions.
- We introduce **RPIembeddor**, a transformer-based model for binary interaction classification and demonstrate its strong performance.

## RNAInterAct

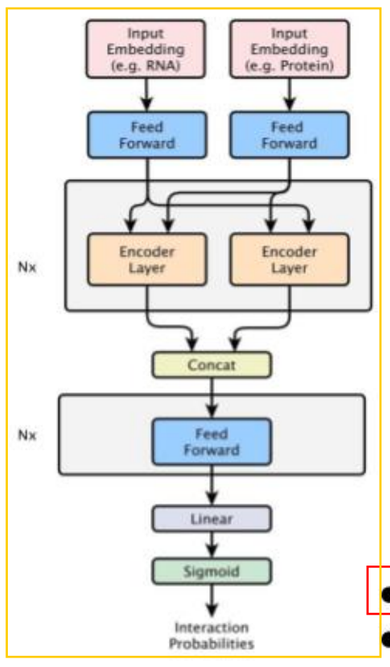
Compilation of the RNAInterAct dataset in five steps:

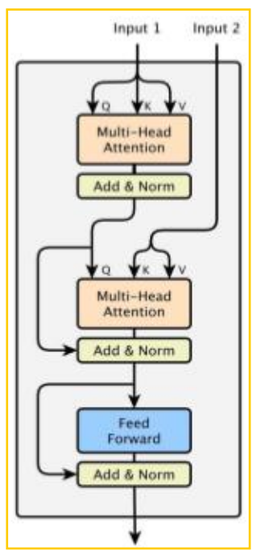
1. **Gathering positive RPIs** from the RNAInter[1] database:
2. Cross-linking large scale databases to **obtain sequence information**.
3. **Filtering**.
4. **Annotating** proteins with clan and RNA with RNA family information.
5. **Generating biologically relevant negative interactions**.

The final dataset contains **122,217 RPIs** with a **1:2 positive to negative ratio**.




## RPIembeddor




- **Protein embeddings:** ESM-2[3].
- **RNA embeddings:** RNA-FM[4].

## RNA family-based train/test split

- Dataset split into **TRinter** (training) and **TSfam** (test) sets **based on RNA families**.
- **Non-homologous data** guarantees evaluations on truly unseen interactions.

| Feature | TRinter | TSfam | RPI2825[2] |
|---|---|---|---|
| Unique RNA Families | 976 | 172 | N/A |
| Unique Protein Clans | 152 | 152 | N/A |
| Positive Interactions | 35,852 | 4,887 | 871 |
| Negative Interactions | 73,362 | 8,116 | 0 |
| Total Interactions | 109,214 | 13,003 | 871 |

## Ablation studies

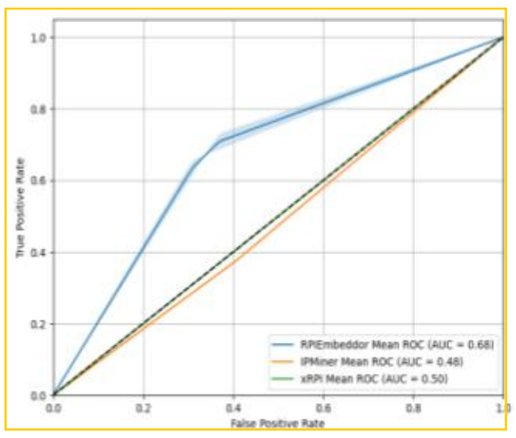
| Model | Prec. | Rec. | F1 | Acc. |
|---|---|---|---|---|
| RPIembeddor | **0.563±0.019** | **0.659±0.071** | **0.605±0.019** | **0.678±0.009** |
| One-Hot | 0.0±0.0 | 0.0±0.0 | 0.0±0.0 | 0.0±0.0 |
| Random-Protein | 0.0±0.0 | 0.0±0.0 | 0.0±0.0 | 0.0±0.0 |
| RNA-Random | 0.0±0.0 | 0.0±0.0 | 0.0±0.0 | 0.0±0.0 |

**Both embeddings contribute to RPIembeddor's performance.**

## Evaluation on TSfam and RPI2825

| Model\Test set | TSfam | | | | RPI2825 | | | |
|---|---|---|---|---|---|---|---|---|
| | Prec. | Rec. | F1 | Acc. | Prec. | Rec. | F1 | Acc. |
| RPIembeddor | **0.550 ±0.010** | 0.627 ±0.017 | **0.586 ±0.013** | **0.667 ±0.009** | **1.0 ±0.0** | 0.667 ±0.085 | 0.8 ±0.049 | 0.667 ±0.085 |
| IPMiner | 0.357 | 0.375 | 0.366 | 0.512 | **1.0** | 0.107 | 0.193 | 0.107 |
| XRPI | 0.375 | **0.909** | 0.531 | 0.398 | 1.0* | 0.982* | 0.991* | 0.982* |

*XRPI has been trained on the RPI2825 dataset.




- **RPIembeddor generalizes across data distributions.**
- **Competitors behave like random classifiers.**

[1] Juanjuan Kang et al. "RNAInter v4.0: RNA interactome repository with redefined confidence scoring system and improved accessibility." Nucleic Acids Research, 2022. doi: 10.1093/nar/gkab997.

[2] Dharm Skandh Jain et al. "A data-driven model for predicting RNA-protein interactions based on gradient boosting machine." *Scientific Reports*, 8(1):9552, 2018.

[3] Z. Lin et al. "Evolutionary-scale prediction of atomic-level protein structure with a language model." *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574. Epub 2023 Mar 16.

[4] Jiayang Chen et al. "Interpretable RNA foundation model from unannotated data for highly accurate RNA structure and function predictions." *arXiv preprint* arXiv:2204.00300, 2022.

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Paper
https://openreview.net/forum?id=04MgOkuPTo

Dataset
https://ml.informatik.uni-freiburg.de/research-artifacts/RNAInterAct/RNAInterAct.tar.gz